ISYE 6650 – Homework 2 (Detailed Solutions)

Question 1

For each of the following, state whether it is discrete or continuous; give the sample space S; and make a reasonable guess for the distribution:

(a) The number of cars that go by until there is a car carrying 3 people

Answer

Discrete random variable.

Explanation

Each car passing is a trial. A "success" occurs when a car with exactly 3 people appears. The random variable counts how many cars pass until the first success.

- Sample space: $S = \{1, 2, 3, \dots\}$
- Distribution: Geometric distribution

(b) The number of cars among the next 20 cars that have more than two bumper stickers

Answer

Discrete random variable.

Explanation

There are exactly 20 trials, each with two outcomes (success/failure).

- Sample space: $S = \{0, 1, 2, \dots, 20\}$
- Distribution: Binomial distribution

(c) Whether or not the next vehicle contains a child

Answer

Discrete random variable.

Explanation

There are two possible outcomes.

- Sample space: $S = \{0, 1\}$
- Distribution: Bernoulli distribution

(d) The number of cars in the next 10 minutes containing two or more children

Answer

Discrete random variable.

Explanation

Counts of events occurring in a fixed time interval.

- Sample space: $S = \{0, 1, 2, \ldots\}$
- Distribution: Poisson distribution

(e) The length of time until the next ambulance

Answer

Continuous random variable.

Explanation

Time is continuous.

- Sample space: $S = (0, \infty)$
- Distribution: Exponential distribution

(f) The total number of people in the next 50 cars

Answer

Discrete random variable.

Explanation

This is a sum of many independent random variables. By the Central Limit Theorem, the distribution is approximately normal.

- Sample space: $S = \{0, 1, 2, \ldots\}$
- Distribution: Approximately Normal

Question 2

Suppose the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0 \\ \frac{1}{2}, & 0 \le b < 1 \\ 1, & 1 \le b < \infty \end{cases}$$

What is the probability mass function of X?
Answer

$$P(X = 0) = \frac{1}{2}, \quad P(X = 1) = \frac{1}{2}$$

Detailed Explanation
The probability mass function (PMF) is obtained by identifying the jumps in the cumulative distribution function (CDF).

Step 1: Identify points of discontinuity

The CDF jumps at:

$$b = 0 \quad \text{and} \quad b = 1$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = F(0) - \lim_{b \to 0^-} F(b)$$

From the definition:

$$F(0) = \frac{1}{2}, \quad \lim_{b \to 0^-} F(b) = 0$$

Therefore:

$$P(X = 0) = \frac{1}{2} - 0 = \frac{1}{2}$$

Step 3: Compute $P(X = 1)$

$$P(X = 1) = F(1) - \lim_{b \to 1^-} F(b)$$

From the definition:

$$F(1) = 1, \quad \lim_{b \to 1^-} F(b) = \frac{1}{2}$$

Thus:

$$P(X = 1) = 1 - \frac{1}{2} = \frac{1}{2}$$

Question 3

Suppose X has a binomial distribution with parameters $n = 6$ and $p = \frac{1}{2}$. Show that $X = 3$ is the most likely outcome.

Answer

$X = 3$ is the most likely value.

Detailed Explanation

Step 1: Write the binomial PMF

$$P(X = k) = \binom{6}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{6-k}$$

Step 2: Simplify the expression

$$P(X = k) = \binom{6}{k} \left(\frac{1}{2}\right)^6$$

Since $\left(\frac{1}{2}\right)^6$ is constant for all k, maximizing $P(X = k)$ reduces to maximizing $\binom{6}{k}$.

Step 3: Evaluate binomial coefficients

$$\binom{6}{0} = 1$$

$$\binom{6}{1} = 6$$

$$\binom{6}{2} = 15$$

$$\binom{6}{3} = 20$$

$$\binom{6}{4} = 15$$

$$\binom{6}{5} = 6$$

$$\binom{6}{6} = 1$$

Step 4: Identify maximum

The maximum value occurs at $k = 3$.

Thus:

$$P(X = 3) \text{ is the largest}$$

Question 4

Let X be a random variable with probability density

$$f(x) = \begin{cases} c(1-x^2), & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find c

Answer

$$c = \frac{3}{4}$$

Detailed Explanation

Step 1: Use total probability rule

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Step 2: Substitute the PDF

$$\int_{-1}^{1} c(1-x^2)\,dx = 1$$

Step 3: Separate the integral

$$c\int_{-1}^{1} 1\,dx - c\int_{-1}^{1} x^2\,dx = 1$$

Step 4: Evaluate each integral

$$\int_{-1}^{1} 1\,dx = 2$$

$$\int_{-1}^{1} x^2\,dx = \left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{2}{3}$$

Step 5: Combine results

$$c\left(2 - \frac{2}{3}\right) = 1$$

$$c \cdot \frac{4}{3} = 1$$

$$c = \frac{3}{4}$$

(b) Find the CDF

Answer

$$F(x) = \begin{cases} 0, & x \leq -1 \\ \frac{1}{2} + \frac{3x}{4} - \frac{x^3}{4}, & -1 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

Detailed Explanation

Step 1: Define the CDF

$$F(x) = \int_{-\infty}^{x} f(t)\, dt$$

Step 2: Substitute PDF

$$F(x) = \int_{-1}^{x} \frac{3}{4}(1 - t^2)\, dt$$

Step 3: Integrate

$$F(x) = \frac{3}{4}\left[t - \frac{t^3}{3}\right]_{-1}^{x}$$

Step 4: Evaluate bounds

$$F(x) = \frac{3}{4}\left(x - \frac{x^3}{3} + \frac{2}{3}\right)$$

Step 5: Simplify

$$F(x) = \frac{1}{2} + \frac{3x}{4} - \frac{x^3}{4}$$

(c) Find the 0.9 quantile

Answer

$$x_{0.9} \approx 0.61$$

Detailed Explanation

Step 1: Set CDF equal to 0.9

$$\frac{1}{2} + \frac{3x}{4} - \frac{x^3}{4} = 0.9$$

Step 2: Multiply by 4

$$2 + 3x - x^3 = 3.6$$

Step 3: Rearrange

$$x^3 - 3x + 1.6 = 0$$

Step 4: Solve numerically

Solving this cubic numerically yields:

$$x \approx 0.61$$

Question 5

Assume monthly demand $X \sim$ Poisson(100). How many items should be ordered so that stockout probability is less than 10%?

Answer

$$Q = 113$$

Detailed Explanation

Step 1: Define stockout probability

$$P(\text{stockout}) = P(X > Q)$$

Step 2: Convert condition

$$P(X > Q) < 0.10 \quad \Longleftrightarrow \quad P(X \leq Q) \geq 0.90$$

Step 3: Express Poisson CDF

$$P(X \leq Q) = \sum_{k=0}^{Q} \frac{e^{-100} 100^k}{k!}$$

Step 4: Use quantile definition

We choose Q such that:

$$F(Q) = 0.90$$

Step 5: Numerical computation

Using a Poisson quantile function:

$$Q = \text{Poisson}_{0.9}(100) \approx 113$$

Thus, ordering 113 items ensures stockout probability is below 10%.